Hybrid CV-DV LCHS

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Abstract

something

Some paper ideas

1. Introduction and Motivation:
   (a) Try to save some ancillary qubits, simplify those scary multi-controlled unitaries, remove truncation and discretization error
   (b) What PDE/ODE system we are interested in? Need a very specific one.
2. Methodology: Around kernel function construction and Hamiltonian evolution oracle design. Kernel function in LCHS: constructed based on CV state preparation and post-selection via CV measurement. Hamiltonian evolution: Hybrid CV-DV oracle.
   (a) How much proof do we need? What theorems in [3, 2, 7] can we directly used for constructing the family of kernel functions?
   (b) What assumptions and conditions are required for the family of our kernel functions?
   (c) Do we want to design new kernel functions or just construct the one with the optimal dependency from existing papers? The kernel function can also be problem specific if there are some ODE/PDE systems we are very interested in.
   (d) Cost for Fock expansion to approximate a desired state
   (e) For Hybrid CV-DV oracle, this may be combined with the implementation part to provide a specific circuit-level implementation. E.g., if there is a scalable way (recursive maybe?) to decompose the matrix coefficients into summation of Pauli stings, and we can use operators in the Continuous LCU paper to do implementation.
3. Implementation and Experiments
   (a) I suppose we just use Bosonic Qiskit to do numerical simulation. But is there some physically limits we need to impose? Like based on the properties of the current hardware, should we need to put some consideration on noise, number of qubits per qumode, etc.?
   (b) How do we fairly compare the resource cost between DV LCHS and CV-DV LCHS? Can we have some time-to-solution estimations?
4. Conclusion and Future Works
5. Check more recent work from Nana Liu on CV for PDE solver. https://iopscience.iop.org/article/10.1088/2058-9565/ad49cf
6. Varational CV-DV ansatz [5]

1 Introduction

2 Motivation

Linear combination of Hamiltonian simulation (LCHS) [3, 2] provides an optimal state preparation cost and a near-optimal scaling in matrix queries on all parameters for solving linear ordinary differential equations (ODEs) in the form of

$$\frac{\mathrm{d}u(t)}{\mathrm{d}t} = -A(t)u(t) + b(t), \quad u(0) = u_0 \tag{1}$$

where $t \in [0, T]$, $A(t) \in \mathbb{C}^{N \times N}$ and $u(t), b(t) \in \mathbb{C}^N$. The solution of the ODE (1) has the explicit form

$$u(T) = \mathcal{T}e^{-\int_0^T A(s)\mathrm{d}s}u_0 + \int_0^T \mathcal{T}e^{-\int_s^T A(s')\mathrm{d}s'}b(s)\mathrm{d}s \tag{2}$$

where $\mathcal{T}$ is the time-ordering operator. To start with, the coefficient matrix $A(t)$ is first decomposed into $A(t) = L(t) + iH(t)$ via $L(t) = \left[A(t) + A^\dagger(t)\right]/2, H(t) = \left[A(t) - A^\dagger(t)\right]/2i$. It is assumed that $L(t) \succeq 0$. Otherwise, we can shift $u(t) = e^{ct}v(t)$ and use $v(t)$ for the ODE system so that $L(t) + cI \succeq 0$, where $-c$ is the minimum of the smallest eigenvalues of $L(t)$. LCHS provides that

$$\mathcal{T}e^{-\int_0^t A(s)\mathrm{d}s} = \int_{\mathbb{R}} g(k)\left[\mathcal{T}e^{-i\int_0^t (kL(s)+H(s))\mathrm{d}s}\right]\mathrm{d}k \tag{3}$$

$$\int_0^t \mathcal{T}e^{-\int_s^t A(s')\mathrm{d}s'}b(s)\mathrm{d}s = \int_0^t \int_{\mathbb{R}} g(k)\left[\mathcal{T}e^{-i\int_s^t (kL(s')+H(s'))\mathrm{d}s'}\right]b(s)\mathrm{d}k\mathrm{d}s \tag{4}$$

where $g(k) := \frac{f(k)}{1-ik}$ for some kernel function $f(k)$ and $t \in [0, T]$. The choice of $f(k)$ that leads to near-optimal complexity dependency on all parameters is

$$f(k) = \frac{1}{(2\pi e^{-2^\beta})e^{(1+ik)^\beta}},$$

where $\beta \in (0, 1)$ is a user-defined constant [2]. Note that the terms in brackets in Eq. (3) and (4) are unitary. Thus, LCHS further truncates and discretizes the integrals in Eq. (3) and (4) into a sum of unitaries, employing state preparation circuits for $|u_0\rangle := u_0/\|u_0\|$ and $|b(s)\rangle := b(s)/\|b(s)\|$ and the linear combination of unitaries (LCU) circuits to prepare the solution state $|u(T)\rangle := u(T)/\|u(T)\|$ [3, 2].

Combining with the finite difference method and linearization methods, for example, Carleman linearization [6], various non-linear partial differential equations can also be converted into a linear ODE system that is solvable by LCHS. However, the number of terms in the discretized LCHS integral, denoted by M, brings a huge cost on quantum resources in the circuit implementation level for either near-term or early fault-tolerant quantum machines. For the complementary solution, M has complexity

$$M \in \mathcal{O}\left(T \max_t \|L(t)\| \left(\log\frac{1}{\epsilon}\right)^{1+1/\beta}\right)$$

where ϵ is the total error incurred by LCHS [2]. The corresponding LCU circuit requires $\lceil \log_2(M) \rceil$ ancillary qubits, and more problematically, $\lceil \log_2(M) \rceil$-controlled multiple-qubit unitary operators for Hamiltonian simulations. In this case, for a time-independent ODE, the value of M can be in the 10^6 magnitude when ϵ is around 10^{-5}, $\beta = 0.8$, $T = 1000$, and $\|L\| = 1$. More exact scalings of LCHS are provided in [7].

Inspired by the continuous LCU in [4], a hybrid CV-DV LCHS is a promising direction to bring down the resource cost and, ideally, remove truncation and discretization errors in the integrals calculation. Specifically, continuous LCU can use $\mathcal{O}(1)$ number of ancillary oscillators, along with CV state preparation and CV measurements, replace $\lceil \log_2(M) \rceil$ ancillary qubits in the DV LCU circuit, while the unitaries for Hamiltonian evolution remain in qubit space. Continuous LCU circuit can compute the integrals in Eq. (3) and (4) without truncation and discretization on integrals, whereas truncations errors may still persist in CV state preparation and hybrid Hamiltonian simulation.

Potential hardware. Trapped-ion devices (Duke Quantum Center?, Pagano Research Group in Rice) and cavity QED devices (Schuster lab in Standford), supporting Gaussian gates (displacement, phase rotation, squeezing) and non-Gaussian (SNAP)

Potential software. Construct a customized backend package for LCHS on HybridLane

3 Methodology

Let us consider a time-independent homogeneous ODE system, where $A(t) := A$ and $b(t) := 0$ for all t. From Eq. (2) and (3), the complementary solution becomes

$$u(t) = \left[\int_{\mathbb{R}} g(k) e^{-it(kL+H)} \mathrm{d}k\right] u_0 \tag{5}$$

for $t \in [0, T]$. Using a single oscillator for the ancillary in continuous LCU, the qubits that synthesize unitary $e^{-it(kL+H)}$ need to couple with this ancillary oscillator, leading to the joint evolution

$$e^{-it(L\otimes\hat{p}+H\otimes I)}$$

where $\hat{p}$ is the momentum operator and I is the identity operator. That is to say, the qumode remains untouched when H acts on the qubit system.

Before forming the circuit for the join evolution, an appropriate function $g(k)$ has to be obtained. To do this, we need to prepare and post-select two appropriate CV states. Let $|\psi\rangle$ be the preparation state, the state prepared in the oscillator before the joint evolution, and $|\phi\rangle$ be the project state, the state post-selected at the end. According to [4], the qumode sandwich gives

$$\langle\phi| e^{-it(L\otimes\hat{p}+H\otimes I)} |\psi\rangle = \int_{\mathbb{R}} \mathrm{d}p\, \phi^*(p)\psi(p) e^{-it(pL+H)}.$$

Compare to the Eq. (5), this gives $g(p) = \phi^*(p)\psi(p)$. Namely, we need to engineer the preparation and post-selection states to obtain an appropriate $g(p)$ that satisfies LCHS's requirements. To simplify the process, the post-selection state $|\phi\rangle$ can be fixed to the vacuum state, so that $g(p) = \phi_0(p)\psi(p)$. Providing $\phi_0(p) = (2/\pi)^{1/4} e^{-p^2}$ [4], we need

$$\psi(p) := \frac{g(p)}{\phi_0(p)} = \frac{g(p)}{(2/\pi)^{1/4} e^{-p^2/\sigma^2}} = f(p)(\pi/2)^{1/4} e^{p^2}(1-ip)^{-1}$$

for some desired $f(p)$ mentioned previously.
We observe that $f(p)$ cannot be arbitrary, since for $\psi(p)$ to be physically realizable, it has to be normalizable, i.e., we want $\int_{\mathbb{R}} |\psi(p)|^2 dp < \infty$. However, for the current kernel function formalism, the quantum state is non-normalizable,

$$\int_{\mathbb{R}} |\psi(p)|^2 dp \sim \int_{\mathbb{R}} \frac{e^{2p^2} dp}{1+p^2} \to \infty$$

Therefore, our desired kernel function will have the following additional property:

$$\int_{\mathbb{R}} |\psi(p)|^2 dp \sim \int_{\mathbb{R}} |f(p)|^2 \frac{e^{2p^2} dp}{1+p^2} < \infty$$

this leads to our initial formalism of a CV/DV LCHS theorem:

Theorem 1 (CV/DV LCHS). *Suppose $A(t) = L(t) + iH(t)$ and $L(t) \succeq 0$, then*

$$\mathcal{T} e^{-\int_0^t A(s)\, ds} = \int_{\mathbb{R}} \frac{f(k)}{1-ik} U_k(t)\, dk.$$

and $f(z)$ is a function of $z \in \mathbb{C}$, such that:

1. ***(Analyticity)*** *$f(z)$ is analytic on the lower half plane $\{z : \mathrm{Im}(z) < 0\}$ and continuous on $\{z : \mathrm{Im}(z) \le 0\}$,*
2. ***(Decay)*** *there exist $\alpha > 0,\ C > 0$ such that $|z|^\alpha |f(z)| \le C$ when $\mathrm{Im}(z) \le 0$,*
3. ***(Normalization)*** *$\int_{\mathbb{R}} \frac{f(k)}{1-ik} dk = 1$.*
4. ***(Square-integrability)*** *$\int_{\mathbb{R}} |f(k)|^2 \frac{e^{2k^2} dk}{1+k^2} < \infty$*

If necessary, the exact desired state $|\psi\rangle$ can be approximated via a finite Fock expansion

$$\psi'(p) \approx \sum_{n=0}^{N_{max}} C_n \langle p|n\rangle$$

where coefficients $\{C_n\}$ are overlaps

$$C_n = \int_{\mathbb{R}} \mathrm{d}p \, \phi_n^*(p)\psi(p)$$

At this point, the missing piece for the whole hybrid CV-DV LCHS circuit is the oracle syntheses for matrices L and H, whose basis-gate implementations are highly dependent on the structures of both matrices and are very problem-specific. Many previous works in the DV regime tackle this problem by decomposing coefficient matrices into linear combinations of Pauli matrices, for example, hyperbolic partial differential equations [8] and advection-diffusion equation [1]. Borrowing their designs and following the circuit synthesis in [4], the evolution of decomposed Hamiltonians coupled with oscillators can be constructed in CV-DV circuits.

3.1 State Preparation

Just expand the optimal kernel function $f(x)$? Or can we engineer a specific one for

3.2 Error bound estimation for kernel function

Theorem 2. *Let*

$$g'(p) = \psi'(p)\,\phi_0(p)$$

denote the truncated approximation of $g(p)$ obtained by restricting the qumode to at most $N_{\max}$ bosonic levels. Then, there exists an integer $N_{\max} \in \mathbb{N}$ such that $\|g'(p) - g(p)\| < \epsilon$, with asymptotic complexity

$$N_{\max} = O\left(\log^2 \frac{1}{\epsilon}\right).$$

Proof. Define the truncated Hilbert space $\mathbb{H}_{N_{\max}} = \mathrm{span}\{\psi_n\}_{n=0}^{N_{\max}}$, and let $\Pi^F_{N_{\max}} : L^2(\mathbb{R}) \to \mathbb{H}_{N_{\max}}$ be the orthogonal projection. Then the projection of ψ is

$$(\Pi^F_{N_{\max}}\psi)(p) = \sum_{n=0}^{N_{\max}} C_n \psi_n(p) = \psi'(p).$$

Thus,

$$\|g(p) - g'(p)\| = \|\phi_0(p)\psi(p) - \phi_0(p)\psi'(p)\| \leq \|\phi_0\|\,\|\psi - \Pi^F_{N_{\max}}\psi\|.$$

TODO:ψ satisfies the following conditions:

1. ψ is analytic in the strip $\mathcal{S}_\rho := \{z \in \mathbb{C} : \Im(z) \in [-\rho, \rho]\}$ for some $\rho > 0$;
2. $\exists \mathcal{K}\, |e^{z^2/2}\psi(z)| \leq \mathcal{K}|z|^\sigma$ as $z \to \infty$ within $\mathcal{S}_\rho$, for some $\sigma \in \mathbb{R}$;
3. The boundary integral $\hat{V} := \int_{\partial \mathcal{S}_\rho} |e^{z^2/2}\psi(z)|\,|dz|$ is finite, where $\partial\mathcal{S}_\rho = \{z \in \mathbb{C} : z = x \pm i\rho,\ x \in (-\infty, \infty)\}$;
4. $N_{\max} \geq \max\{\lfloor \sigma \rfloor + 1, 0\}$.

By the convergence analysis of Hermite approximations [9], under the stated analyticity and growth assumptions, the error satisfies

$$\|\psi - \Pi^F_{N_{\max}}\psi\| \leq \mathcal{K} e^{-\rho\sqrt{2N_{\max}}}.$$

Hence, choosing $N_{\max} = O(\log^2(1/\epsilon))$ suffices to guarantee $\|g' - g\| < \epsilon$. □

3.3 Error bound dependence on kernel function

The strict error estimation parameters $\sigma, \rho, \mathcal{K}$ will depend on the kernel function $g(p)$.

3.4 Oracle Synthesis

3.4.1 Example: Classical Harmonic Oscillator

Consider position $x(t)$, mass m, angular frequency ω, the Harmonic oscillator is modelled as

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = -\omega^2 x, \tag{6}$$

which translate to

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix} x \\ v \end{bmatrix} = -\begin{bmatrix} 0 & -1 \\ \omega^2 & 0 \end{bmatrix}\begin{bmatrix} x \\ v \end{bmatrix} \tag{7}$$

and define the coefficient matrix as A. We need to do Cartesian decomposition

$$L = \frac{A + A^\dagger}{2} = \frac{1}{2}\left(\omega^2 - 1\right) X \tag{8}$$

$$H = \frac{A - A^\dagger}{2i} = \frac{1}{2}\left(-\omega^2 - 1\right) Y \tag{9}$$

Thus, in LCHS circuit, the unitaries are

$$\exp(-iTkL) = \exp\left(-iX\frac{Tk}{2}\left(\omega^2 - 1\right)\right) \tag{10}$$

$$\exp(-iTH) = \exp\left(-iY\frac{T}{2}\left(-\omega^2 - 1\right)\right) \tag{11}$$

Thus, the Hamiltonian simulation can be easily done with RX and RY gates (need some trotterization on time steps).

3.4.2 Example: Discretized One-dimensional Heat Equation

Consider one-dimensional (1D) heat equation

$$\frac{\partial u(x,t)}{\partial t} = \alpha\frac{\partial^2 u(x,t)}{\partial x^2} \tag{12}$$

where α (or often written in a squared constant) is called the thermal diffusivity of a length-L rod. Its initial condition is

$$u(x,0) = f(x), 0 < x < L. \tag{13}$$

And we start with the easiest boundary condition, a homogeneous Dirichlet boundary condition

$$u(0,t) = u(L,t) = 0, t > 0. \tag{14}$$

With a 6-point (2 end-points, x_0 and x_5 are fixed by the boundary condition) second order central difference approximation

$$\left.\frac{\partial^2 u(x,t)}{\partial x^2}\right|_{x_j} \approx \frac{u(x_{j+1}) - 2u(x_j) + u(x_{j-1})}{h^2} \tag{15}$$

we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix} u(x_1,t) \\ u(x_2,t) \\ u(x_3,t) \\ u(x_4,t) \end{bmatrix} = -\frac{1}{h^2}\begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}\begin{bmatrix} u(x_1,t) \\ u(x_2,t) \\ u(x_3,t) \\ u(x_4,t) \end{bmatrix}, u(x,0) = f(x) \tag{16}$$

And to decompose the coefficient matrix into Pauli words,

$$T = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} = 2I_0I_1 - I_0X_1 - \frac{1}{2}\left(X_0X_1 + Y_0Y_1\right) \tag{17}$$

And $A = 1/h^2T$. Since A is real symmetric,

$$L = \frac{A + A^\dagger}{2} = A \tag{18}$$

$$H = \frac{A - A^\dagger}{2i} = \mathbf{0} \tag{19}$$

3.4.3 Example: Two dimensional Heat Equation

Two dimensional heat equation

$$\frac{\partial u(x,y,t)}{\partial t} = \alpha\left(\frac{\partial^2 u(x,y,t)}{\partial x^2} + \frac{\partial^2 u(x,y,t)}{\partial y^2}\right), u(x,y,0) = f(x,y) \tag{20}$$

for $0 < x < a$ and $0 < y < b$. With homogeneous Dirichlet boundary conditions

$$u(0,y,t) = u(a,y,t) = 0 \le y \le b, t \ge 0 \tag{21}$$

$$u(x,0,t) = u(x,b,t) = 0 \le x \le a, t \ge 0 \tag{22}$$

Again, using central difference method, we have

$$\frac{\mathrm{d}u_{i,j}(t)}{\mathrm{d}t} \approx \frac{u_{i+1,j}(t) - 2u_{i,j}(t) + u_{i-1,j}(t)}{h_x^2} + \frac{u_{i,j+1}(t) - 2u_{i,j}(t) + u_{i,j-1}(t)}{h_y^2} \tag{23}$$

where $u_{i,j}(t) = u(x_i, y_j, t)$.

Let N be the number of interior points (boundary points are determined by boundary condition)

$$N = \lceil a/h_x - 1\rceil \cdot \lceil b/h_y - 1\rceil. \tag{24}$$

Suppose there 2 interior points in x-direction and 4 interior points in y-direction, we denote

$$u_1(t) = u(x_1, y_1, t), u_2(t) = u(x_1, y_2, t), \dots \tag{25}$$

$$u_5(t) = u(x_2, y_1, t), u_6(t) = u(x_2, y_2, t), \dots \tag{26}$$

Note that in this scheme, the neighbor point in y direction is ± 1 indices, but neighbor in x direction is $\pm\lceil b/h_y - 1\rceil$ indices.

For the form of $u_t = -Au(t)$, in our ordering scheme

$$A_{k,k} = \frac{2}{h_x^2} + \frac{2}{h_y^2} \tag{27}$$

And the neighbor in x direction is on the super and sub diagonal

$$A_{k\pm1,k} = -\frac{1}{h_y^2} \tag{28}$$

And the neighbor in y direction is

$$A_{k,k\pm4} = -\frac{1}{h_x^2}. \tag{29}$$

An example coefficient matrix is

$$A = \begin{pmatrix}
\frac{2}{h_x^2}+\frac{2}{h_y^2} & -\frac{1}{h_y^2} & 0 & 0 & -\frac{1}{h_x^2} & 0 & 0 & 0 \\
-\frac{1}{h_y^2} & \frac{2}{h_x^2}+\frac{2}{h_y^2} & -\frac{1}{h_y^2} & 0 & 0 & -\frac{1}{h_x^2} & 0 & 0 \\
0 & -\frac{1}{h_y^2} & \frac{2}{h_x^2}+\frac{2}{h_y^2} & -\frac{1}{h_y^2} & 0 & 0 & -\frac{1}{h_x^2} & 0 \\
0 & 0 & -\frac{1}{h_y^2} & \frac{2}{h_x^2}+\frac{2}{h_y^2} & 0 & 0 & 0 & -\frac{1}{h_x^2} \\
-\frac{1}{h_x^2} & 0 & 0 & 0 & \frac{2}{h_x^2}+\frac{2}{h_y^2} & -\frac{1}{h_y^2} & 0 & 0 \\
0 & -\frac{1}{h_x^2} & 0 & 0 & -\frac{1}{h_y^2} & \frac{2}{h_x^2}+\frac{2}{h_y^2} & -\frac{1}{h_y^2} & 0 \\
0 & 0 & -\frac{1}{h_x^2} & 0 & 0 & -\frac{1}{h_y^2} & \frac{2}{h_x^2}+\frac{2}{h_y^2} & -\frac{1}{h_y^2} \\
0 & 0 & 0 & -\frac{1}{h_x^2} & 0 & 0 & -\frac{1}{h_y^2} & \frac{2}{h_x^2}+\frac{2}{h_y^2}
\end{pmatrix}.$$

And the decompose is

$$\begin{aligned}
A &= \frac{1}{h_y^2} I \otimes T + \frac{2}{h_x^2} I_0 I_1 I_2 - \frac{1}{h_x^2} X_0 I_1 I_2 && (30)\\
&= \frac{1}{h_y^2}\left[2I_0I_1I_2 - I_0I_1X_2 - \frac{1}{2}\left(I_0X_1X_2 + I_0Y_1Y_2\right)\right] + \frac{2}{h_x^2} I_0I_1I_2 - \frac{1}{h_x^2} X_0I_1I_2 && (31)\\
&= \left(\frac{2}{h_y^2} + \frac{2}{h_x^2}\right) I_0I_1I_2 - \frac{1}{h_y^2}\left(I_0I_1X_2 - \frac{1}{2}\left(I_0X_1X_2 + I_0Y_1Y_2\right)\right) - \frac{1}{h_x^2} X_0I_1I_2 && (32)
\end{aligned}$$

4 Implementations

5 Experiments

6 Conclusion

7 Future Works

8 Author Contributions

If needed

9 Acknowledgments

10 Competing interests

All authors declare no financial or non-financial competing interests.

11 Data and code availability

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Appendix

Some LCHS basics if needed

Complementary (Homogeneous) Solution $\mathcal{T}e^{-\int_0^t A(s)\mathrm{d}s}u_0$

Remark 1. *With a quadrature rule, the goal is to find the approximation*

$$\int_a^b g(x) \approx \sum_{j=0}^{m-1} w_j g(x_j). \tag{33}$$

For an integral over $[-1,1]$ with Gaussian quadrature with m nodes, set node x_j to be the root of Legendre polynomial of degree m, $P_m(x)$ with weight $w_j = \frac{2}{(1-x_j^2)((P_m'(x_j))^2}$. For a longer interval $[a,b]$, the composite rule is applied by dividing the interval $[a,b]$ into K segments, i.e., $[a+kh, a+(k+1)h]$ where $h = \frac{b-a}{K}$ is the step size. Gaussian quadrature is applied on each shorter interval and it is constructed from that on $[-1,1]$ by a change of variable. That is, the weight $w_j \to \frac{h}{2}w_j$ and node $x_j \to \frac{h}{2}x_j + (a+kh) + \frac{1}{2}h$.

Consider the ODE solution over a truncated finite interval $[-K, K]$ using the composite Gaussian quadrature to discretize the interval.

The first step is to define

$$g(k) = \frac{1}{C_\beta(1-ik)e^{(1+ik)^\beta}}, \quad U(T,k) = \mathcal{T}e^{-i\int_0^T (kL(s)+H(s))\mathrm{d}s}. \tag{34}$$

Then,

$$\begin{aligned}
\mathcal{T}e^{-\int_0^T A(s)\mathrm{d}s} &= \int_{\mathbb{R}} g(k)U(T,k)\mathrm{d}k && \text{(by LCHS)} \\
&\approx \int_{-K}^{K} g(k)U(T,k)\mathrm{d}k && \text{(truncation)} \\
&= \sum_{m=-K/h_1}^{K/h_1-1} \int_{mh_1}^{(m+1)h_1} g(k)U(T,k)\mathrm{d}k && \text{(divide to shorter intervals)} \\
&\approx \sum_{m=-K/h_1}^{K/h_1-1} \sum_{q=0}^{Q-1} c_{q,m}U(T,k_{q,m}) && \text{(Gaussian quadrature in each interval)} \quad (35) \\
&= \sum_{j=0}^{M-1} c_j U(T,k_j) && \text{(simplified notation)} \quad (36)
\end{aligned}$$

where h_1 is the step size in the composite Gaussian quadrature, Q is the number of nodes in each interval $[mh_1, (m+1)h_1]$, $k_{q,m}$ is a Gaussian node, $c_{q,m} = w_q g(k_{q,m})$, w_q is a Gaussian weight (which does not depend on the choice of m), and M is the overall number of the nodes (or unitaries).

Assumption 1. *Assume K and h_1 are chosen so that K/h_1 is an integer.*

To bound the truncation error by $\epsilon > 0$, it suffices to choose

$$K = \mathcal{O}\left((\log(1/\epsilon))^{1/\beta}\right)$$

To bound the quadrature error by $\epsilon > 0$, it suffices to choose

$$h_1 = \frac{1}{eT\max_t \|L(t)\|}, \quad Q = \left\lceil \frac{1}{\log 4}\log\left(\frac{8}{3C_\beta}\frac{K}{\epsilon}\right)\right\rceil = \mathcal{O}\left(\log\frac{1}{\epsilon}\right), \tag{37}$$

and the overall number of unitaries in the summation formula is

$$M = \frac{2KQ}{h_1} = \mathcal{O}\left(T\max_t \|L(t)\| \left(\log\frac{1}{\epsilon}\right)^{1+1/\beta}\right) \tag{38}$$

For the specific value of K, note that in Eq. (176) in [2], we have

$$\left\| \int_{\mathbb{R}} g(k)U(T,k)\mathrm{d}k - \int_{-K}^{K} g(k)U(T,k)\mathrm{d}k \right\| \leq \frac{2^{B+1}B!}{C_\beta(\cos(\beta\pi/2))^B}\frac{1}{K}e^{-\frac{1}{2}K^\beta\cos(\beta\pi/2)} \leq \epsilon$$

$$\ln\left(\frac{2^{B+1}B!}{C_\beta(\cos(\beta\pi/2))^B}\right) + \ln(1/\epsilon) \leq \ln(K) + \frac{1}{2}K^\beta\cos(\beta\pi/2) \tag{39}$$

$$2\cdot\frac{\ln\left(\frac{2^{B+1}B!}{C_\beta(\cos(\beta\pi/2))^B}\right) + \ln(1/\epsilon) - \ln(K)}{\cos(\beta\pi/2)} \leq K^\beta \tag{40}$$

where $B = \lceil 1/\beta \rceil$

Inhomogeneous Term $\int_0^t \mathcal{T}e^{-\int_s^t A(s')\mathrm{d}s'}b(s)\mathrm{d}s$

The discretization of the inhomogeneous term is

$$\begin{aligned}
&\int_0^T \mathcal{T}e^{-\int_s^T A(s')\mathrm{d}s'}b(s)\mathrm{d}s \\
&= \int_0^T \int_{\mathbb{R}} \frac{f(k)}{1-ik}U(T,s,k)b(s)\mathrm{d}k\mathrm{d}s && \text{(by LCHS Eq. (??))} \\
&\approx \int_0^T \sum_{m_1=-K/h_1}^{K/h_1-1}\sum_{q_1=0}^{Q_1-1} c_{q_1,m_1}U(T,s,k_{q_1,m_1})b(s)\mathrm{d}s && \text{(by Eq. (35))} \\
&\approx \sum_{m_2=0}^{T/h_2-1}\sum_{q_2=0}^{Q_2-1}\sum_{m_1=-K/h_1}^{K/h_1-1}\sum_{1_1=0}^{Q_1-1} c'_{q_2,m_2}c_{q_1,m_1}U(T,s_{q_2,m_2},k_{q_1,m_1})\,|b(s_{q_2,m_2})\rangle && \text{(composite Gaussian quadrature)} \quad (41) \\
&= \sum_{j'=0}^{M'-1}\sum_{j=0}^{M-1} c'_{j'}c_jU(T,s_{j'},k_j)\,|b(s_{j'})\rangle && \text{(simplified notation)} \quad (42)
\end{aligned}$$

where $U(T,s,k) = \mathcal{T}e^{-i\int_s^T(kL(s')+H(s')\mathrm{d}s'}$, h_2 is the step size for the discretization of the time integral, Q_2 is the number of Gaussian quadrature nodes, s_{q_2,m_2} is a node, $c'_{q_2,m_2} = w'_{q_2}\|b(s_{q_2,m_2})\|$, the w'_{q_2} is a Gaussian weight. The $|b\rangle$ implies that it is the normalized vector $b/\|b\|$.

Let

- $\Lambda = \sup_{p\geq 0, t\in[0,T]} \|A^{(p)}\|^{1/(p+1)}$
- $\Xi = \sup_{p\geq 0, t\in[0,T]} \|b^{(p)}\|^{1/(p+1)}$, where (p) refers to the p^{th}-order time derivative
- $\|b\|_{L^1} = \int_0^T \|b(s)\|\mathrm{d}s$

To bound the approximation error of Eq. (42) by $\epsilon > 0$, it suffices to choose

$$K = \mathcal{O}\left(\left(\log\left(1+\frac{\|b\|_{L^1}}{\epsilon}\right)\right)^{1/\beta}\right),\ h_1 = \frac{1}{eT\max_t\|L(t)\|}, h_2 = \frac{1}{eK(\Lambda+\Xi)}, \tag{43}$$

$$Q_1 = \mathcal{O}\left(\log\left(1+\frac{\|b\|_{L^1}}{\epsilon}\right)\right), Q_2 = \mathcal{O}\left(\log\left(\frac{T(\Lambda+\Xi)}{\epsilon}\right)\right) \tag{44}$$

and

$$M' = \frac{TQ_2}{h_2} = \mathcal{O}\left(TK(\Lambda+\Xi)\log\left(\frac{T(\Lambda+\Xi)}{\epsilon}\right)\right) = \tilde{\mathcal{O}}\left(T(\Lambda+\Xi)\left(\log\left(1+\frac{\|b\|_{L^1}}{\epsilon}\right)\right)^{1/\beta}\log(1/\epsilon)\right)$$

As indicated in Eq. (234) and (235) in [2], we have

$$\|c\|_1\frac{eT(\Lambda+\Xi)}{4^{Q_2}} \leq \frac{\epsilon}{2}$$

$$\frac{1}{\ln(4)}\ln\left(\|c\|_1\frac{2eT(\Lambda+\Xi)}{\epsilon}\right) \leq Q_2, \tag{45}$$

where $\|c\|_1 = \mathcal{O}(1)$. For computing Q_1, just use the same formula in Eq. (37) but with the K in Eq. (43).

Summary

With LCHS, Eq. (2) becomes

$$u(T) \approx \sum_{j=0}^{M-1} c_j U(T,k_j) |u_0\rangle + \sum_{j'=0}^{M'-1} \sum_{j=0}^{M-1} c'_{j'} c_j U(T, s_{j'}, k_j) |b(s_{j'})\rangle . \tag{46}$$

where $U(T,s,k) = \mathcal{T} e^{-i\int_s^T (kL(s')+H(s'))\mathrm{d}s'}$ and $U(T,k) = U(T,0,k)$. So, we need oracles circuit to implement the Hamiltonian simulation operator $U(T,s,k)$ and oracles to encode the coefficients c and c' into amplitudes. Then, use LCU to obtain $u(T)$.

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